

Muhammad Tahir

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EDUCATION

Universiti Malaya, Kuala Lumpur

MSCS – Research (AI/ML Focus)

March 2026 – Present

FAST-NUCES, Karachi

BS Computer Science

August 2021 – June 2025

CGPA: 3.45 | Ranked 43rd / 398 students

CERTIFICATIONS & AWARDS

- Rector's List Award (Spring 2025)
- Dean's List Awards (Fall 2023, Spring 2024, Fall 2024)
- Data Cleaning in Python – Datacamp (2024)
- Docker – KodeKloud (2024)
- Data Analysis with Python – Coursera (2023)
- AI for Everyone – Coursera (2023)

LANGUAGE PROFICIENCY

- **English:** IELTS 7.0 (No band below 6.0)
- **Urdu:** Native

TECHNICAL SKILLS

Languages & Frameworks

React / React Native	TypeScript
Next.js / NestJS	Node.js / Express
Spring Boot	Python / FastAPI
Java	PHP
C/C++	HTML/CSS

AI / ML

PyTorch	Agentic AI
RAG Pipelines	Medical Imaging

Tools & Infrastructure

Docker	AWS (EC2, S3)
Git / GitHub	Ubuntu / Bash
Cypress	JUnit 5
CUDA / MPI / OpenMP	SPSS

Databases

PostgreSQL	MongoDB
MySQL	

SOFT SKILLS

- Strong communication and collaboration
- Critical thinking and problem-solving
- Effective time management and multitasking

INTERESTS

- AI/ML and Healthcare Technology
- Full-Stack Development & Open Source
- Startups and Entrepreneurship
- Fitness and Sports

EXPERIENCE

Ikhwa Technologies

Jan 2025 – Apr 2026 | Associate Software Engineer

- Led full-stack development of **Aviation Index**, a subscription-based aviation intelligence platform, delivering 10+ production features using React, TypeScript, NestJS, and PostgreSQL.
- Engineered and deployed analytics dashboards, airport mapping tools, subscription workflows, and advertising modules used by aviation professionals globally.
- Enhanced **TrackOps** by implementing custom document generation, reporting workflows, resource-based ACL, and third-party API integrations, reducing manual operational efforts by writing end-to-end test cases with Cypress.

Bloomrix

Aug 2024 – Nov 2024 | Software Engineering Intern

- Contributed to the delivery of multiple client projects using React.js, Node.js, and PHP in Agile development environments.
- Collaborated with cross-functional teams to implement new features, resolve defects, and improve application performance.

FAST-NUCES, Karachi

Spring 2025 | TA – Computer Architecture (EE3009)

- Assisted in evaluating assignments, quizzes, and project submissions; recognised as responsible and hardworking by department faculty.

RESEARCH & SELECTED PROJECTS

Multi-Agent System for Cardiac MRI Analysis

March 2026 – Present | MSCS Research, Universiti Malaya

- Developing a five-agent pipeline (Router, Ingestion, Preprocessing, Classification, Decision Support) for automated Hypertrophic Cardiomyopathy detection from CMR slices.
- Implementing 3D ResNet-18 for volumetric classification on the HCM Kaggle dataset; integrating RAG-based clinical guidelines into diagnostic decision support.

Neurological Disease Identification via Deep Learning

May 2025 | Final Year Research, FAST-NUCES

- Designed a CNN-based system achieving 75% accuracy in identifying Alzheimer's disease-associated SNPs from genomic data.
- Introduced the Phenotype Influence Score (PIS), outperforming XGBoost and Random Forest baselines; applied to the ADNI dataset identifying APOE and novel risk loci.

Ebre Rider · iOS/Android (Google Play & App Store)

- Developed and deployed a cross-platform passenger transport application serving real users on Android and iOS.
- Built mobile applications using React Native and developed a business operations dashboard using React.js and Spring Boot.

Restaurant Food Ordering Platform · Web Application

- Designed and developed a full-stack food ordering platform with secure authentication and payment integration using Next.js, React.js, TypeScript, Tailwind CSS, Node.js, and MongoDB.
- Integrated Google OAuth, Facebook OAuth, and Stripe Payment API to provide a seamless user experience.